

1.1. Let $\lambda = |CA| : |CB|$ be the ratio in which the point $C \in [AB]$ divides the segment $[AB]$. Check that

$$\overrightarrow{OC} = \frac{1}{1+\lambda} \overrightarrow{OA} + \frac{\lambda}{1+\lambda} \overrightarrow{OB}.$$

1.2. Consider the parallelograms $A_1A_2A_3A_4$ and $B_1B_2B_3B_4$. Show that the midpoints of the segments $[A_1B_1]$, $[A_2B_2]$, $[A_3B_3]$ and $[A_4B_4]$ are the vertices of a parallelogram.

1.3. In each of the following cases, decide if the indicated vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ can be represented with the vertices of a triangle:

a) $\mathbf{u}(7, 3), \mathbf{v}(2, 8), \mathbf{w}(-5, 5)$

b) $\mathbf{u}(1, 2, -1), \mathbf{v}(2, -1, 0), \mathbf{w}(1, -3, 1)$

1.4. Let $\mathcal{K} = (O, \mathcal{B})$ and $\mathcal{K}' = (O', \mathcal{B}')$ be two frames in $\mathbb{E}^2$, with $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ and $\mathcal{B}' = (\mathbf{i}', \mathbf{j}')$. Assume that $O', \mathbf{i}'$ and $\mathbf{j}'$ are known relative to $\mathcal{K}$:

$$[O']_{\mathcal{K}} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{B}} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{B}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Determine the base change matrix from $\mathcal{K}$ to $\mathcal{K}'$ and the coordinates with respect to $\mathcal{K}'$ of the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

1.5. Consider the tetrahedron $ABCD$ and the frames

$$\mathcal{K}_A = (A, \mathcal{B}) \text{ with } \mathcal{B} = (\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}),$$

$$\mathcal{K}_B = (B, \mathcal{B}) \text{ with } \mathcal{B} = (\overrightarrow{BA}, \overrightarrow{BC}, \overrightarrow{BD}),$$

$$\mathcal{K}'_A = (A, \mathcal{B}) \text{ with } \mathcal{B} = (\overrightarrow{BA}, \overrightarrow{BD}, \overrightarrow{BC}).$$

Determine

a) the coordinates of the vertices of the tetrahedron in the three coordinate systems,

b) the base change matrices from $\mathcal{K}'_A$ to $\mathcal{K}_A$ and from $\mathcal{K}_A$ to $\mathcal{K}_B$,

c) the relative orientation of the three frames.

1.6. Let $\mathcal{K} = (O, \mathcal{B})$ and $\mathcal{K}' = (O', \mathcal{B}')$ be two frames in $\mathbb{E}^3$ with $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$ and $\mathcal{B}' = (\mathbf{i}', \mathbf{j}', \mathbf{k}')$. Assume that $O', \mathbf{i}', \mathbf{j}'$ and $\mathbf{k}'$ are known relative to $\mathcal{K}$:

$$[O']_{\mathcal{K}} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{i}' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{j}' = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{k}' = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Determine the coordinates with respect to $\mathcal{K}'$ of the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \quad [D]_{\mathcal{K}} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}.$$

1.7. What is the number of Cartesian frames that one can construct with the vertices of a tetrahedron?